Probability

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I. INTRODUCTION

II. RANDOM EVENTS

Given an *elementary event* with a countable set of discrete random outcomes, $E_1, E_2, \ldots, E_n, \ldots$, there is associated with each possible outcome E_k a number called a *probability*, p_k, which can lie between 0 and 1, $0 \leq p_k \leq 1$.

If the kth outcome never occurs, $p_k = 0$; if it is sure to occur, $p_k = 1$. Conversely, if $p_k = 0$, we say that the event almost surely does not occur; and if $p_k = 1$, the event almost surely occurs. Another notation for the probability of event E_k is $P\{E_k\} = p_k$.

The following are some simple properties of the probability of events:

1. $P\{E_i \text{ and/or } E_j\} \leq p_i + p_j$.

2. E_i and E_j are said to be *mutually exclusive* events if and only if the occurrence of E_i implies that E_j does not occur, $E_i \implies \bar{E}_j$ (not E_j), and vice versa. If E_i and E_j are mutually exclusive, $P\{E_i \text{ and } E_j\} = 0, P\{E_i \text{ or } E_j\} = p_i + p_j$. 3. A whole class of events can be mutually exclusive for all i and j. When the class is *exhaustive*, that is, all possible events have been enumerated, $P\{\text{some } E_i\} = \sum_i p_i = 1$.

   In the following, we consider a compound experiment consisting of just two elementary events. For clarity, we imagine the first to have outcomes $\{E_i\}$ with probability p_{1i} and the second to have outcomes $\{F_j\}$ and probabilities p_{2j}, respectively. Each of p_{1j} and p_{2j} obeys statements 1, 2, and 3 above. An outcome of such a composite event is a pair (E_i, F_j).

4. The probability of the specific outcome (E_i, F_j) is p_{ij}, called the *joint probability* for E_i and F_j.

5. $p_{ij} = p_{1i} \cdot p_{2j}$, if and only if events E_i and F_j are *independent*.

6. Suppose E_i and F_j are not independent; then the joint probability can be written as

$$p_{ij} = \left(\sum_k p_{ik}\right)\left(\frac{p_{ij}}{\sum_k p_{ik}}\right) = p(i)\left(\frac{p_{ij}}{\sum_k p_{ik}}\right), \tag{2.1}$$

   $p(i)$ defines a new number called the *marginal probability* for event E_i, that is, the probability that E_i does, in fact, occur, whatever the second event may be. Therefore,

$$\sum_i p(i) = \sum_i \sum_k p_{ik} = 1 \tag{2.2}$$

   The marginal distribution for the second event, F_j, can similarly be written as $p(j) = \sum_k p_{kj}$.

7. The second factor of Eq. (2.1) is the *conditional probability*

$$p(j|i) = \frac{p_{ij}}{\sum_k p_{ik}} \tag{2.3}$$

   and is the probability for event F_j occurring, given that event E_i has occurred. The probability for some F_j should be 1, and indeed

$$\sum_j p(j|i) = \sum_j \frac{p_{ij}}{\sum_k p_{ik}} = 1 \quad \text{for every } i. \tag{2.4}$$

All joint probabilities can be factored into a marginal distribution and a conditional probability. This scheme can be generalized to treat the joint occurrence of three or more elementary events.

III. RANDOM VARIABLES

In the following discussion we shall assume that for every elementary outcome E_i, there is an associated real number x_i. A random selection, X, of one of the possible values $x_1, x_2, \ldots$ is called a random variable. The probability that the value x_i is chosen is given by $p_i = P\{X = x_i\}$. The *expectation* of this random variable X, that is, the stochastic mean value, is defined as

$$E(X) \equiv \sum_i P\{X = x_i\}x_i = \sum_i p_i x_i = \mu. \tag{3.1}$$

μ is called the *expected* or *mean value*. It is common in physics to write this as $\langle X \rangle$, and we shall often use that notation.

Consider some real-valued function $g(x_i) = g_i$, where the x_i correspond to a countable set of elementary events with probabilities p_i. If X is a random variable, then $g(X)$ is also a random variable. The expectation of $g(X)$ is defined as

$$E(g(X)) = \langle g(X) \rangle = \sum_i p_i g(x_i). \tag{3.2}$$

From the definition of the expected value of a function, we have the property that

$$\langle \text{constant} \rangle = \text{constant}, \tag{3.3}$$

and that for any constants λ_1, λ_2 and two functions g_1, g_2,

$$\langle \lambda_1 g_1(X) + \lambda_2 g_2(X) \rangle = \lambda_1 \langle g_1(X) \rangle + \lambda_2 \langle g_2(X) \rangle. \tag{3.4}$$

An important application of expected values is to the powers of X. The nth moment of X is defined as the expectation of the nth power of X,

$$\langle X^n \rangle \equiv \sum_i p_i x_i^n, \tag{3.5}$$

The *central moments* of X are given by

$$\langle (X - \mu)^n \rangle \equiv \sum_i p_i (x_i - \langle X \rangle)^n. \tag{3.6}$$

The second central moment has particular significance,

$$\langle (X - \mu)^2 \rangle = \langle (X - \langle X \rangle)^2 \rangle = \sum_i p_i (x_i - \langle X \rangle)^2 = \sum_i p_1 x_i^2 - \langle X \rangle^2 = \langle X^2 \rangle - \langle X \rangle^2, \tag{3.7}$$

and is called the *variance* of X or $\text{var}\{X\}$. The square root of the variance is a measure of the dispersion of the random variable. It is referred to as the *standard deviation* and sometimes the *standard error*. The variance of a function of the random variable, $g(X)$, can be determined as

$$\text{var}\{g(X)\} = \langle (g(X) - \langle g(X) \rangle)^2 \rangle = \langle g^2(X) \rangle - \langle g(X) \rangle^2. \tag{3.8}$$

Consider two random variables, X and Y. They are are not in general independent. Two random variables are said to be independent if they derive from independent events. As we have seen in Eq. (3.4), the expectation of a linear combination is the linear combination of the expectations. This result does not require that X and Y be independent. The effect of statistical dependence will be seen in the variance of a linear combination of the two variables,

$$\text{var}\{\lambda_1 X + \lambda_2 Y\} = \lambda_1^2 \text{var}\{X\} + \lambda_2^2 \text{var}\{Y\} + 2[\lambda_1 \lambda_2 \langle XY \rangle - \lambda_1 \lambda_2 \langle X \rangle \langle Y \rangle]. \tag{3.9}$$

The expectation of the product is

$$\langle XY \rangle = \sum_{i,j} p_{ij} x_i y_j. \tag{3.10}$$

If X and Y are independent, $p_{ij} = p_{1i} \cdot p_{2j}$ and

$$\langle XY \rangle = \sum_i p_{1i} x_i \sum_j p_{2j} y_j = \langle X \rangle \langle Y \rangle. \tag{3.11}$$

Then Eq. (3.9) becomes simply

$$\mathrm{var}\{\lambda_1 X + \lambda_2 Y\} = \lambda_1^2 \mathrm{var}\{X\} + \lambda_2^2 \mathrm{var}\{Y\}. \tag{3.12}$$

When X and Y are not necessarily independent, we introduce a new quantity, the *covariance*, which is a measure of the degree of independence of the two random variables X and Y:

$$\mathrm{cov}\{X, Y\} = \langle XY \rangle - \langle X \rangle \langle Y \rangle. \tag{3.13}$$

The covariance equals 0 when X and Y are independent and $\mathrm{cov}\{X, X\} = \mathrm{var}\{X\}$.

Note that zero covariance does not by itself imply independence of the random variables. The following simple example illustrates that even functional dependence can still yield a zero covariance. Let X be a random variable that may be -1, 0, or 1 with equal probabilities, and define $Y = X^2$. Obviously, $\langle X \rangle = 0, \langle XY \rangle = \langle X^3 \rangle = 0$ so $\mathrm{cov}\{XY\} = 0$.

The covariance can have either a positive or negative value. Another quantity derived from the covariance is the correlation coefficient,

$$\rho(X, Y) = \frac{\mathrm{cov}\{X, Y\}}{\sqrt{\mathrm{var}\{X\}\mathrm{var}\{Y\}}}, \tag{3.14}$$

so that $-1 \leq \rho(X, Y) \leq 1$. Since the covariance can be positive or negative, the variance of a linear combination of two dependent random variables can be greater or less than the variance if the variables were independent.

A. The Binomial Distribution

Consider two events E_0 and E_1 that are mutually exclusive and exhaustive:

$$P\{E_1\} = p, \quad x_1 = 1, \tag{3.15}$$
$$P\{E_0\} = 1 - p, \quad x_0 = 0. \tag{3.16}$$

Let X_i be the random variable that is the ith outcome of a series of N such events. The expectations of X_i and its square become

$$E(X_i) = p \cdot 1 + (1 - p) \cdot 0 = p, \tag{3.17}$$
$$E(X_i^2) = p, \tag{3.18}$$

and the variance is then

$$\mathrm{var}\{X_i\} = p - p^2 = p(1 - p). \tag{3.19}$$

Each outcome is either 0 or 1, and we set X to be the sum of the N outcomes $\{X_i\}$,

$$X = \sum_{i=1}^{N} X_i. \tag{3.20}$$

The probability that $X = n$ is the probability that n of the X_i were 1 and $N - n$ were 0. That is,

$$P\{X = n\} = \binom{N}{n} p^n (1 - p)^{N-n} \quad n = 0, 1, \ldots, N. \tag{3.21}$$

This is the *binomial distribution*. $\binom{N}{n} = N!/n!(N - n)!$ is the binomial coefficient, which counts the number of different ways in which the n E_1 and the $(N - n)$ E_0 may occur. The expected value of X is

$$\langle X \rangle = \sum_{n=0}^{N} n \binom{N}{n} p^n (1 - p)^{N-n} = Np. \tag{3.22}$$

This may be verified by direct computation of the sum or by noting that the expected value of X is the sum of the expected value of all the X_i. The variance of X is easily determined; since the X_i are independent, the result in Eq. (3.12) may be employed and

$$\langle (X - Np)^2 \rangle = \sum_{i=1}^{N} \mathrm{var}\{X_i\} = \sum_{i=1}^{N} p(1 - p) = Np(1 - p). \tag{3.23}$$

B. The Geometric Distribution

Suppose we carry out a certain experiment repeatedly and independently, where there are only two outcomes: failure or success. If the outcome is a failure, the experiment will be repeated; otherwise, we stop the procedure. Let the random variable X of interest be the number of times we repeat the experiment until the first success. Let q be the probability of failure in one experiment and $p = 1 - q$ be the probability of success. Then

$$P\{X = n\} = q^{n-1}p, \quad n = 1, 2, \ldots. \tag{3.24}$$

The average number of experiments that will be carried out is

$$\langle X \rangle = \sum_{n=1}^{\infty} nq^{n-1}p = \frac{p}{(1-q)^2} = \frac{1}{p}. \tag{3.25}$$

The variance of X can be calculated as

$$\mathrm{var}\{X\} = \langle X^2 \rangle - \langle X \rangle^2 = \left(\frac{2}{p^2} - \frac{1}{p}\right) - \frac{1}{p^2}. \tag{3.26}$$

As an example, in particle transport problems, the number of collisions one particle makes follows this distribution if the medium is infinite, homogeneous, and if the relative probability of different outcomes is constant.

C. The Poisson Distribution

A random variable X is said to follow a *Poisson distribution* when

$$P\{X = n\} = \frac{\lambda^n}{n!}e^{-\lambda}, \quad n = 0, 1, \ldots, \tag{3.27}$$

where λ is a parameter of the distribution. It is easy to show that

$$\langle X \rangle = \lambda, \tag{3.28}$$

$$\mathrm{var}\{X\} = \lambda. \tag{3.29}$$

This distribution is fundamental in the theory of probability and stochastic processes. It is of great use in applications such as research into queuing service systems and similar discrete event problems.

IV. CONTINUOUS RANDOM VARIABLES

V. EXPECTATIONS OF CONTINUOUS RANDOM VARIABLES

VI. BIVARIATE CONTINUOUS RANDOM DISTRIBUTIONS

VII. SUM OF RANDOM VARIABLES

VIII. DISTRIBUTION OF THE MEAN OF A RANDOM VARIABLE

IX. DISTRIBUTION OF THE SUM OF INDEPENDENT RANDOM VARIABLE

X. MONTE CARLO INTEGRATION

XI. MARKOV PROCESSES

We will now furnish concrete, analytical expressions for the probabilities characterizing three important Markov processes. We begin with the so-called Wiener process.

A. Wiener Process

The *Wiener process*, described by $\omega(t)$ for $t > 0$, is characterized by the probability distributions

$$p(\omega_0, t_0) = \frac{1}{\sqrt{4\pi D t_0}} e^{-\omega_0^2/4Dt_0}, \tag{11.1}$$

$$p(\omega_1, t_1|\omega_0, t_0) = \frac{1}{\sqrt{4\pi D \Delta t}} e^{-(\Delta\omega)^2/4D\Delta t}, \tag{11.2}$$

$$\Delta\omega = \omega_1 - \omega_0, \tag{11.3}$$

$$\Delta t = t_1 - t_0. \tag{11.4}$$

The probabilities are parameterized through the constant D, referred to as the diffusion constant, since the probability distributions $p(\omega_0, t_0)$ and $p(\omega_1, t_1|\omega_0, t_0)$ are solutions of the diffusion equation (**??**) discussed extensively below. The Wiener process is homogeneous in time and space, which implies that the conditional transition probability $p(\omega_1, t_1|\omega_0, t_0)$ depends only on the relative variables $\Delta\omega$ and Δt. Put differently, the probability $p(\Delta\omega, \Delta t)$ for an increment $\Delta\omega$ to occur is independent of the current state of the Wiener process $\omega(t)$. The probability $p(\Delta\omega, \Delta t) = p(\omega_0 + \Delta\omega, t_0 + \Delta t|\omega_0, t_0)$ is given by Eq. (11.2).

Characteristic Functions, Moments, Correlation Functions and Cumulants for the Wiener process

In case of the Wiener process simple expressions can be provided for the characteristics introduced above, i.e., for the characteristic function, moments and cumulants. Combining (11.2) and (**??**) one obtains for the characteristic function

B. Ornstein-Uhlenbeck Process

Our second example for a Markov process is the *Ornstein-Uhlenbeck process*. The Ornstein-Uhlenbeck process, describing a random variable $v(t)$, is defined through the probabilities

$$p(v_0, t_0) = \frac{1}{\sqrt{\pi\gamma\sigma^2}} e^{-v_0^2/\gamma\sigma^2}, \tag{11.5}$$

$$p(v_1, t_1|v_0, t_0) = \frac{1}{\sqrt{\pi S}} e^{-\left(v_1 - v_0 e^{-\gamma\Delta t}\right)^2/S}, \tag{11.6}$$

$$S = \gamma\sigma^2\left(1 - e^{-2\gamma\Delta t}\right), \tag{11.7}$$

$$\Delta t = |t_1 - t_0|. \tag{11.8}$$

The probabilities are characterized through two parameters σ and γ. Their significance will be explained further below. The process is homogeneous in time, since (11.6) depends solely on t, but is not homogeneous in v. Furthermore, the Ornstein-Uhlenbeck Process is stationary, i.e., $p(v_0, t_0)$ does not change in time.

Characteristic Functions, Moments, Correlation Functions and Cumulants for the Ornstein-Uhlenbeck process

C. White Noise Process

An important idealized stochastic process is the so-called "*Gaussian white noise*". This process, denoted by $\xi(t)$, is not characterized through conditional and unconditional probabilities, but through the following statistical moment and correlation function

$$\langle \xi(t) \rangle = 0, \tag{11.9}$$

$$\langle \xi(t_1)\xi(t_0) \rangle = \zeta^2 \delta(t_1 - t_0). \tag{11.10}$$

The attribute *Gaussian* implies that all cumulants higher than of second order are 0

$$\left\langle\!\left\langle \xi^{n1}(t_1)\xi^{n0}(t_0) \right\rangle\!\right\rangle = \begin{cases} \zeta^2\delta(t_1 - t_0) & \text{for } n_0, n_1 = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{11.11}$$

The reason why this process is termed *white* is connected with its correlation function (11.10), the Fourier transform of which is constant, i.e., entails all frequencies with equal amplitude just as white radiation. The importance of the process $\xi(t)$ stems from the fact that many other stochastic processes are described through stochastic differential equations with a (white) noise term $\xi(t)$. We will show this for the Wiener process below and for the Ornstein-Uhlenbeck processes later in the notes.

As hinted by the examples in this section we can show that the integral of Gaussian white noise happens to be a Wiener process. We prove this in the same fashion as above by deriving the cumulants for $\int dt\xi(t)$. Again the task is simplified by the fact that only one cumulant is non-zero, namely,

$$\begin{aligned}\left\langle\!\!\left\langle \int_0^{t_1} ds_1\ \xi(s_1) \int_0^{t_0} ds_0\ \xi(s_0) \right\rangle\!\!\right\rangle &= \int_0^{t_1} ds_1 \int_0^{t_0} ds_0\, \langle\!\langle \xi^{n1}(t_1)\xi^{n0}(t_0) \rangle\!\rangle \\ &= \int_0^{t_1} ds_1 \int_0^{t_0} ds_0\ \zeta^2\delta(t_1 - t_0) \\ &= \zeta^2 \min(t_1, t_0). \end{aligned} \tag{11.12}$$

We demonstrated, thereby, the important relationship between white noise $\xi(t)$ and the Wiener process $\omega(t)$

$$\omega(t) = \int_0^t ds\ \xi(s), \tag{11.13}$$

for $2D = \zeta^2 = 1$.

XII. STOCHASTIC DIFFERENTIAL EQUATIONS

XIII. ITO AND STRATONOVICH CALCULUS

XIV. EINSTEIN DIFFUSION EQUATION

XV. SMOLUCHOWSKI DIFFUSION EQUATION

XVI. ADJOINT SMOLUCHOWSKI DIFFUSION EQUATION

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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